Real-Time Bike-Sharing Availability & Regional Performance Analytics

1. Project Overview and Objective

- This project focuses on the transformation of raw, flat-file operational data from a multi-city bike-sharing system into a structured Star Schema data model. The primary goal is to provide actionable insights into bike availability, station health, and regional performance trends.
- **Key Objectives**:
  - Data Modelling: Convert a single flat table into one central Fact table and two Dimension tables to improve reporting efficiency.
  - Operational Monitoring: Categorize stations based on real-time stock levels (e.g., Critical Empty, Low Stock, Healthy) to identify service gaps.
  - Geospatial Analysis: Utilize Latitude and Longitude coordinates to visualize station distribution across different contract regions.
  - Trend Tracking: Implement Year-to-Date (YTD) and Monthly analysis to understand seasonal shifts in bike usage.

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2. Data Sources

The dataset consists of bike station statuses from various regions in France.

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3. Problem Statement

- The bike-sharing system generates vast amounts of raw operational data, but it is currently stored in a "flat" format that makes it difficult to perform deep analysis or track historical performance.
- The specific challenges addressed in this project include:
  - **Lack of Actionable Categorization**: Raw numbers of "Available Bikes" do not immediately tell a manager which stations are failing. We need a logic-based system to flag "Critical" stations.
  - **Inefficient Data Structure**: Storing repetitive station addresses and features alongside fluctuating bike counts increases file size and slows down reporting.
  - **Time-Series Complexity**: The raw Last Update timestamps are not formatted for standard "Year-to-Date" or monthly comparison, making it hard to see if the system is improving over time.

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4. Attribute (Column/Features) Details

Attribute Name	Data Type	Description
Number	Whole Number	ID to identify Bike Stations
Total Bike Stands	Whole Number	Maximum No of Slots available at the Station
Available Bike Stands	Whole Number	Count of empty Slots
Available Bikes	Whole Number	Count of bikes currently at the Station
Last Update	Date/Time/Timezone	Timestamp
Stock Status	Text	Category
Occupancy Rate	Percentage	Percentage to show how full the station is

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5. Tools & Technologies

- **Excel**: Converted raw data obtained into Fact and Dimension Tables.
- **Power BI:** Data cleaning & modelling, DAX calculations, visualization, and interactive dashboard creation.

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6. Data Pre-Processing (Power Query)

Tasks Performed:

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, and created calculated fields.
- **Filtering & Sorting:** Organized data to focus on relevant records.

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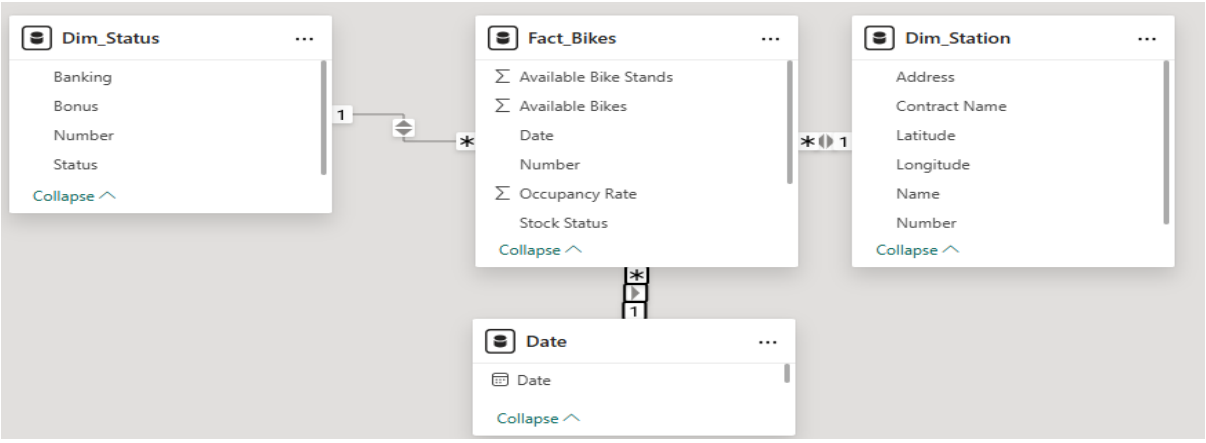
7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.



- **DAX Measures:** Implemented DAX formulas for key metrics, such as Station Count, Average Bikes, Bikes YTD, Occupancy%, Priority Rank, Total Available Bikes, Full Station Count, & Total Capacity.

Station_Count	Avg_Bikes	Bikes_YTD	Occupancy%	Priority_Rank
1,028	7	19,646	34.34%	625

Total_Available_Bikes	Full_Station_Count	Total_Capacity
19,646	1,028	57,203

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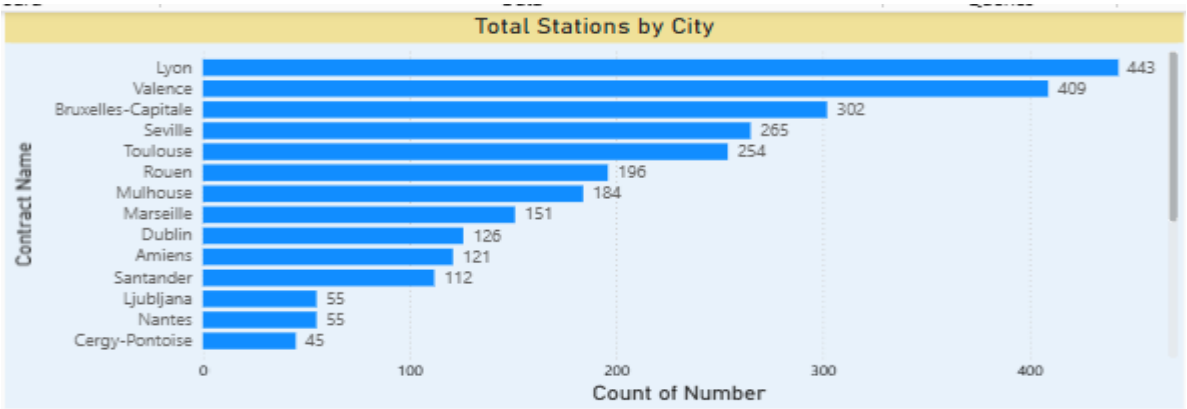
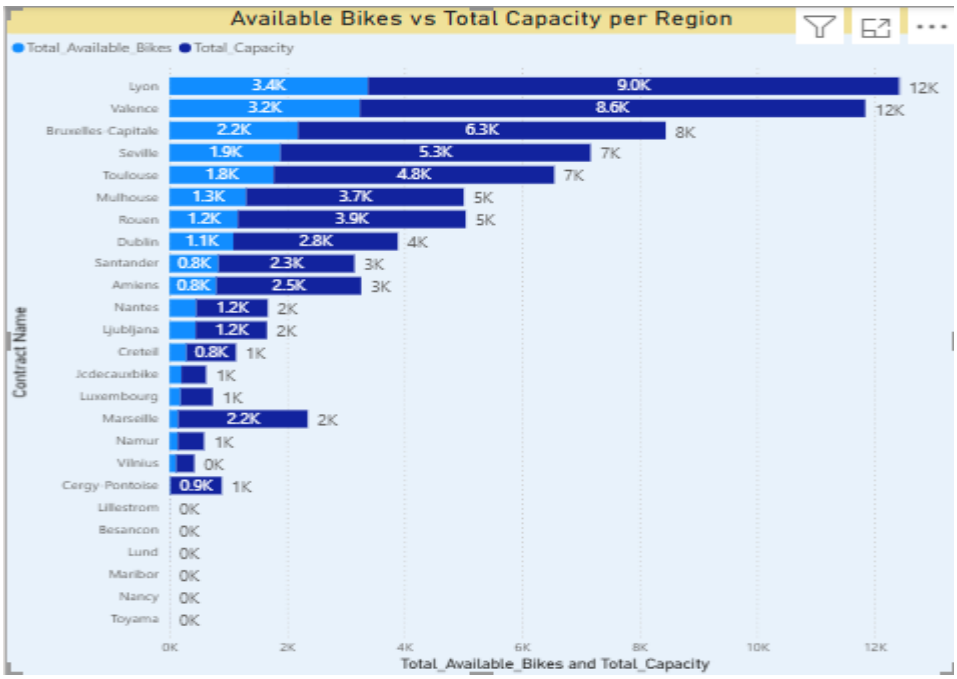
8. Insights & Conclusions

- **Key Findings:**
  - **Infrastructure Lead**: Lyon and Valence are the largest hubs, with Lyon alone containing 443 stations and a total capacity of 9,000 stands.
  - **Critical Supply Gaps**: A high frequency of "Critical Empty" flags exists in Marseille, where all 129 stations currently show zero available bikes.
  - **Operational Saturation**: Major cities like Lyon (419 "Critical Full" stations) are reaching maximum capacity, suggesting a need for physical stand expansion in high-traffic zones.
  - **Data Quality Alert**: The system contains non-operational "Test" and "Warehouse" records that must be excluded to maintain the accuracy of your real-time redistribution audits.
  - **Growth Potential**: While the current forecast shows a flat trend of 20,000 available bikes, the infrastructure in cities like Nancy and Vilnius is underutilized due to a total lack of inventory.
- **Analysis Insights:**
- **Current Distribution of Bike Stations across Different Cities**: Lyon & Valence are the largest operational hubs.



- **Total number of Available Bikes compared to Total Stands for each city**: Lyon & Valence shows there are approximately 3 available bikes for every 9-12 stands. Several cities, including Vilnius, Lillestrom, & Nancy show 0 available bikes despite having total stand capacity.

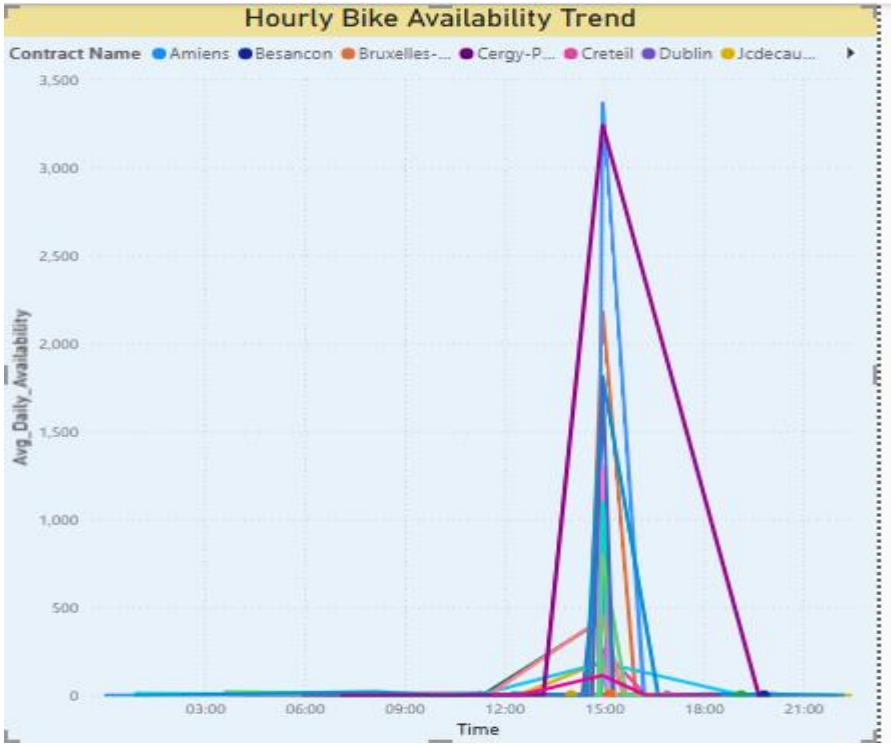


- **Regions having higher frequency of "Critical Empty" stations compared to others**: Marseille has 129 stations flagged as "Critical Empty". In contrast, regions like Lyon have 419 stations in this category, while smaller regions like Amiens have only 8. Larger cities naturally have more stations, so they will often have a higher number of empty stations. However, when a city like Marseille has 100% of its stations (129 out of 129) showing as "Critical Empty," it indicates a regional system outage or a significant redistribution failure rather than just high demand.

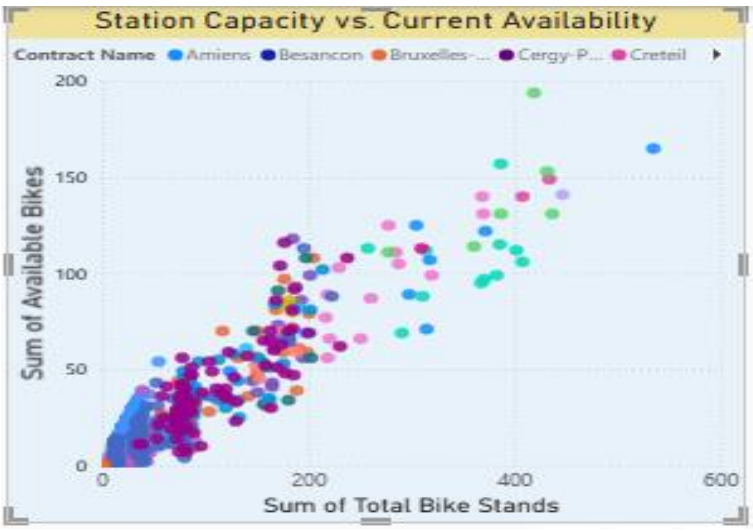
Station Status Distribution by Region

Contract Name	Critical Empty	Critical Full	Healthy	Low Stock	Near Full	**Total**
Amiens	8	8	8	8	8	**8**
Besancon	1	1	1	1	1	**1**
Bruxelles-Capitale	83	83	83	83	83	**83**
Cergy-Pontoise	45	45	45	45	45	**45**
Creteil	2	2	2	2	2	**2**
Dublin	17	17	17	17	17	**17**
Jcdecauxbike	9	9	9	9	9	**9**
Lillestrom	1	1	1	1	1	**1**
Ljubljana	7	7	7	7	7	**7**
Lund	1	1	1	1	1	**1**
Luxembourg	5	5	5	5	5	**5**
Lyon	419	419	419	419	419	**419**
Maribor	1	1	1	1	1	**1**
Marseille	129	129	129	129	129	**129**
Mulhouse	14	14	14	14	14	**14**
Namur	1	1	1	1	1	**1**
Nancy	1	1	1	1	1	**1**
Nantes	9	9	9	9	9	**9**
Rouen	11	11	11	11	11	**11**
Santander	6	6	6	6	6	**6**
Seville	55	55	55	55	55	**55**
Toulouse	122	122	122	122	122	**122**
Toyama	1	1	1	1	1	**1**
Valence	79	79	79	79	79	**79**
Vilnius	1	1	1	1	1	**1**
Total	**1028**	**1028**	**1028**	**1028**	**1028**	**1028**

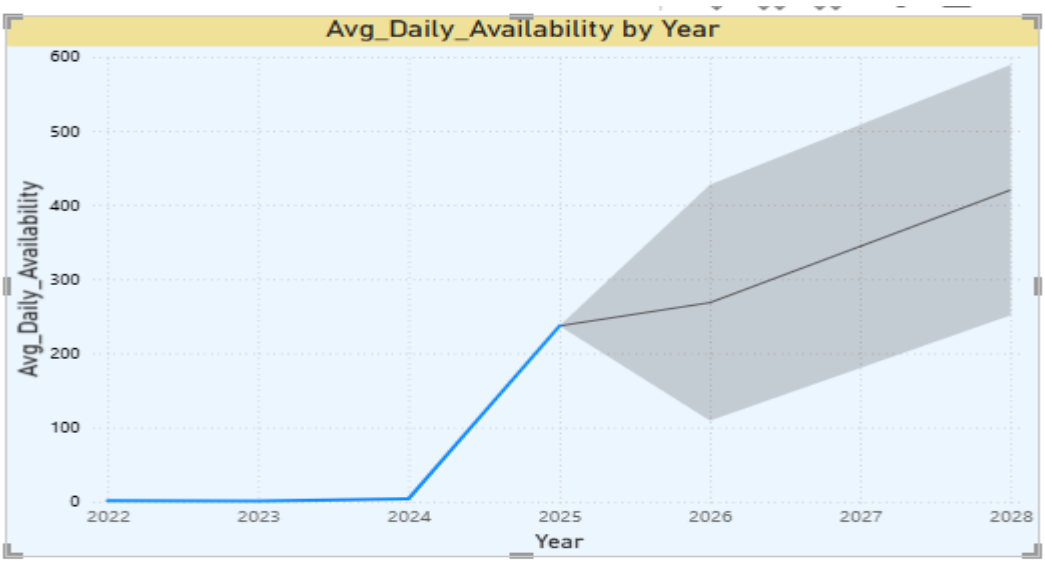
- **Specific Time of day when Bike Availability Drops Significantly in High-Traffic Areas**: Availability does not actually "drop" during a typical morning or evening rush; instead, the chart shows a massive surge in bike availability peaking around 15:00 (3:00 PM). In high-traffic areas like Lyon and Dublin, availability remains near zero for most of the day until this sudden afternoon spike, where it jumps to over 3,000 available bikes globally. Availability is at its lowest - meaning the system is most utilized or "Empty" - during the early morning hours (00:00 to 09:00) and again in the late evening after 18:00.



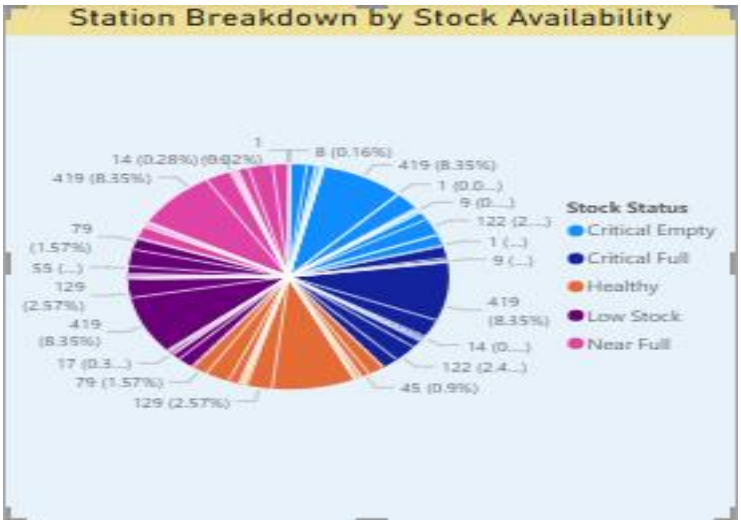
- **Predicted "Critical Full" Stations**: Stations represented by dots in the top-right corner, where the Sum of Available Bikes is almost equal to the Sum of Total Bike Stands, are predicted to become "Critical Full" within the hour. Specific stations, particularly those approaching the 400-stand capacity mark with high bike counts, require immediate attention to free up docking space for arriving commuters.



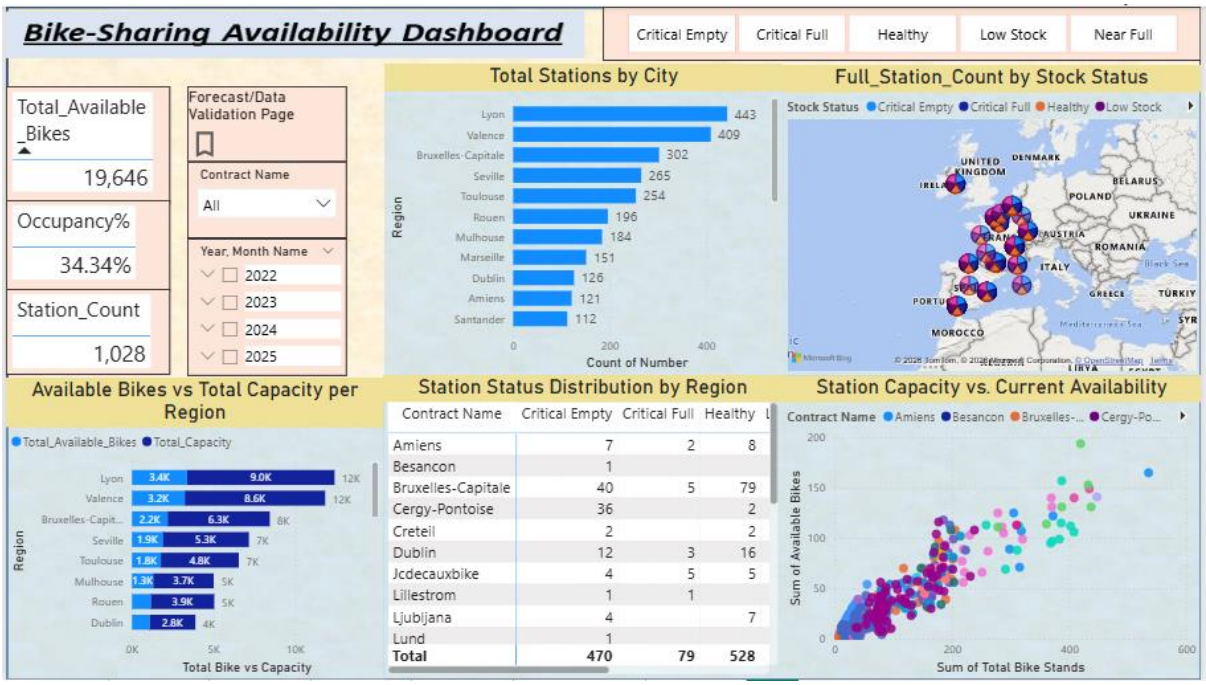
- **Expected Bike Demand:** Upper bound of 427 and lower bound of 109, indicates positive raise in demand.



- **5 stations we should prioritize for bike redistribution to minimize "Critical Empty" flags**: Marseille Central Hubs: With 129 stations all sitting at "Critical Empty," this entire city is your highest priority. Lyon High-Traffic Zones: Lyon has 419 stations in "Critical Empty" status, which is the largest volume of empty stations in your entire dataset. Nancy Stations: This city shows 0 available bikes on your inventory chart, meaning every station there is currently failing to serve users. Vilnius Stations: Like Nancy, this region has 0 available bikes despite having physical capacity, requiring an immediate "seed" of new inventory. Lillestrom Stations: This is the third region showing a total lack of available bikes (0K), placing it in the top tier for urgent redistribution.
- **Addressing "Critical Full" Stations through Capacity Expansion**: We should prioritize increasing bike stands in Lyon and Valence. While these cities already have the highest capacity, they also show the highest volume of bikes, meaning they are "magnets" for riders and frequently hit the "Critical Full" threshold during peak hours.



9. Consolidated Report/Dashboard



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10. Conclusions

To ensure the continued efficiency and growth of the bike-sharing network, it is recommended to prioritize immediate redistribution of stock to "Critical Empty" hubs in Marseille and Lyon to restore service availability. Long-term infrastructure planning should focus on expanding physical stand capacity in high-demand regions like Valence and Lyon to prevent recurring "Critical Full" lockouts. Additionally, the system's data integrity must be maintained by permanently filtering out "Test" and "Warehouse" records from operational dashboards to ensure redistribution trucks are dispatched only to active public locations. Finally, implementing more granular time-based tracking will allow the system to better predict and prepare for peak commuter shifts, ensuring a balanced and reliable user experience across all cities.